

ActInf Textbook Group ~ Cohort 5 ~ Session 4 (Chapter 2, Part 1) ~ 10/10/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 4 (Chapter 2, Part 1) ~ 10/10/2023 | 56:49

all right welcome back everyone it is uh
October 10th 23 and we're in our first
discussion on chapter
two so before we jump into any questions
or
anything what is anyone's just takeaway
from chapter 2 or what were they curious
about or what something that kind of
remains with them after reading feel
free to just raise your hand hand or
just go for
it yeah hi it's Pablo here
um in the page 35 action has a pic value
and pragmatic uh value um brings Rewards
so that for me for what I'm building is
super important so yeah I I have also
some other notes but and a question over
there we can cover later so yeah for me
this is
like my pick on
this awesome yes the idea that we don't
have to coers epistemic value or
learning or Information Gain into
utility we don't have to like convert it
into utility in order for that policy to
be
selected key
aspect any other favorite quotes or fun
parts of chapter two
or parts they woke up sweating about
or I for me tying you tying this back to
the surprise and and and um maybe
elaborating on what encompasses policy I
I'm I'm putting strategy in there but I
don't know if that's correct

specifically in the low
road cool like policy
enumerates the possibilities for Action
over a given time Horizon but strategy
isn't just
enumerating branching path
so then what is strategy great
questions any other just random chapter
two
thoughts you know when it comes to this
idea of planning as inference something
that I've been just kind of thinking
about a lot lately is that um ultimately
when I do something and the best example
is just trying to walk somewhere all I
really do is to hold that conscious
intention to be somewhere and then then
I'm kind of out of the loop other than
having uh you know processing um sensory
evidence that I'm actually getting there
and as long as long as that's happening
you know I I can sort of see this this
Loop in my in my brain where I'm going
and just enjoying myself looking at
things and just thinking all kinds of
other thoughts and so on so forth and
the going kind of happens on its own all
all I really do is to have a conscious
intention I don't even necessarily have
to feel my body as it's moving if I get
into this state almost like a you know
like a walking meditation then it really
feels like you know being in a
self-driving car so it's kind of
interesting that um you know I'm
starting by understanding all of this
stuff like having a having a generative
model to notice these things in my own

awareness I'm starting to notice these things in my own awareness and I'm starting to be you know not only uh have have these ideas confirmed mathematically but also exponentially so just a little bit weird but um really fun

awesome cool and and also no amount of propositional intention to walk you can think as hard as you want I want to walk but then actually the intention that gets you out of the chair is something different so what does that what does that mean or what does that have to do with action yeah and but but but so you have to you have to sort of um distinguish different kinds of intention so like there's this there's the intention as in kind of like it would be nice if or like wishful thinking I I I wish I could do this um but what I mean by intention is a very specific mental state and that mental state is usually not available to you unless you start training in meditation where it becomes obvious that it exists so it's something that like you know if if you think about like think about this in from the perspective of training neural network for robot to do something so you know robot will have some kind of goal in a neural network getting to a place or whatever so it must it must be some kind of activation in the neural network that's that's going to be triggering a sequence of other activations right or a bunch of activations that are going to go through some kind of cycle in time

that that sort of thing and normally
that's an unconscious thing but I I
don't I don't think it has to it has to
be so and like the Buddhist meditative
practices particularly inside meditation
they specifically you know distinguish
the intention so at some point you might
be become aware like first they have an
intention to move the hand then they
have a feeling of moving the hand so as
the I'm holding the intention
and I'm getting the sensor input to
confirm that what I intended is
happening and that process is actually
uh it's possible to introspect it's
possible to to see it in your own
awareness and that's um that's that's
wonderful

really

awesome any

other opening thoughts from people just
some low road or chapter two thought or
something that they're excited about to
explore a little bit today

um the um the getting a little deeper
into what is learning versus
inference yeah multiple time

scales that computational models of
cognition play out on the fastest
sometimes being called inference the
kind of Rapid Fire neural spiking type
little bit slower attention with a

Precision waiting stepping on the gas
pulling off the gas and then a little
bit slower even

learning sometimes the whole process is
called inference sometimes inference is
only being used to refer to the kind of

like crack of the whip most rapid time
scale but what is learning the higher
order phenomena like child learning how
to read how do we connect that with a
more narrow statistical concept of
learning like we learned that the average
height of the tree in the forest was 10
feet but previously we thought it was
nine so that's

learning so is that the same as belief
updating what would anyone else say
what's what is the relationship between
learning and belief

updating I'm curious

about um if uh this system is uh
deterministic or it's more on the will
uh Free Will side

and or or nonone at

[Laughter]

all are you say saying about the belief
updating or about the active inference
about the active inference

okay you know I I like to joke that free
will you know you have you have free
will only so far as that you're free to
suffer if you really understand that
you're not a thing that you're a process
and you're totally interdependent with
everything else then you know the um the
best thing to do is always to you know
be a free energy minimizing agent to do
it in a in an optimal way if you're not
doing that in an optimal way that's
cause that will cause some friction or
suffering we can jump to this uh cohort
five question that's kind
of related um and then
yeah go who was

that okay um sorry oh yeah Ali go for it
uh okay so uh the relation between
belief updating and learning is that
belief updating is much more General
than just learning so uh both inference
and learning um involve belief updating
in some sense and uh this is uh one of
the probably most confusing things about
active inference modeling because most
people when it comes to learning they
only think of belief updating uh as
apply as applied to Only The Learning
side of active inference uh but uh then
basian belief updating uh can be applied
to both inference side and uh the
learning side so they use the exact same
kind of belief basian belief updating
mechanism uh so uh we'll see how Works
uh in relation to both more explicitly
chapter

six yeah the faster inference scale is
kind of like you have the calculator
preloaded like the weights in the neuron
Network are fixed if you want to think
of it that way or the parameters are
kind of like pre-loaded and preset and
then inference is like new argument
comes in new input comes in and it just
it's like a plug-in chug that's the
faster um inferential scale which can
still result in a belief updating and
then learning is actually the change on
those weights or like the change in
that in that system how it's set up so
then influences future rounds of fast
inference
um so in that respect um it's almost
like what Kristoff was saying uh Visa

incent ality prior to a generative model
where learning is changing those
parameters you know prior to having the
generative model being able to update
that

intentionality which the which then the
belief updating happens in relation
to perhaps learning could be like what
kind of thing am I and then the faster
inference is like given the thing I am
how do I

respond and then the kind of General
theme is instead of evaluating sense
making or decision making in terms of
like maximally rewarding their cast in
terms of most

likely so instead of given what I value
what's the most

rewarding it's like given the generative
model what's the most likely
outcome and then in chapter nine we'll
look at at given what empirically did
happen what's the most likely generative
model so that's the kind of scientific
like behavioral researcher going from
the data to the most likely model and
then the other direction is from the
model generating the most likely data
and both of them are are um they're just
two two directions on the same
freeway um

all right let's look at this question
and then anyone else can like add a
question to the table or or
um write another question in the um chat
okay this means that Free Will is part
of the model question Point 2.3 the
interplay of top down and bottom up

processes distinguishes the inferential
view from alternative approaches that
only consider bottomup
processes okay probably a lot that um
could be explored here what what what
does anyone want to kind of bring in
here or say about this
section yeah that that was my question
so yeah my thought was
um if it's Boton up it would be determin
everything is around

U

it's um getting like

uh

um you are just like uh observing
and just getting the data in and you
probably won't have much um action on on
that or your agency I think we call
it and if you is top down it means that
I have an agency I'm deciding I'm I'm
modeling the the world that surrounds
me and

um yeah that was my my question my
thought and something that I
care it's very pH philosophical as well
I think and and

yeah anyone have a thought on
this yeah this is Andrew um I when I
hear this question I I just want to
recommend Eric hell's new book um I I
forget what it's called right now um
sorry I'm multitasking as usual um but
he he talks about how those uh I think
active in inference um is completely
congruent with his theory of um causal
emergence which talks about how you can
have bottom up processes and top down
and and still have some room for free

will in between um with just uh
mechanistic um uh rule following behavior
and and so I feel like there's room for
um the philosophy to still exist Within
a um a a uh
non-dualist uh approach to um
metaphysics yeah thank you Andrew yeah
yeah thank you that makes
sense if I may add a little bit um you
know I I think it's interesting to to
consider here um sort of Steven wal's
principle of computational equivalence
so um W from based on his experience
with computation says that essentially
um there is a level in the computational
ladder ladder of sophistication and it's
actually pretty low it's it's incredibly
low that um once you hit it you're as
sophisticated computationally as any
other agent so even if you had a
completely computationally deterministic
system where everything is fixed like
the universe there is absolutely no way
you're going to be able to properly
predict um out compute um even the
simplest agents like simplest cellular
autom and so on because they are
effectively as computationally
sophisticated as you are so that gives
always rise to um
uncertainty it's it's a building feature
that uh you will not be able to compute
the perfectly a model of internal States
and this is comes up in active inference
modeling the difficulty of constructing
these generative models but it seems a
fundamental principle of nature that
that's impossible so as an agent you

know um if you're as an agent you're
kind of a controller for future State
you have an intention that you want to
achieve a future State and you're trying
trying to sort of get there but the
world that you're trying to control is
never going going to be um simply
deterministically following rules so
there's going to be resolution of some
uncertainties and I think that
resolution of uncertainties that gives
the rise to the feeling of of free will
the underlying system might be
completely deterministic but the
experience of it from from within an
um a part of it embedded in it in the
system it's always we can to feel that I
had these choices because I always had
some
uncertainty
yeah yeah yeah that that's another thing
that I Steve W from it's I I always
wonder how those these two currents are
going to cross
over yeah yeah I suspect on a machine
learning Street talk with Carl and
Steven in the coming months we'll see
though that should be quite um
interesting um Oli and then
Susan uh okay so when it comes to agency
I suppose active
inference kind of assumes that any
lifelike property including uh agency
and um many other things uh they're kind
of uh inherent to uh and uh also their
emerging property of any dynamical
non-equil Dre system uh and most of more
often than not these systems are erotic

and also they possess markup blankets so
um uh in other in other words F supposes
organisms have this uh goal of keeping
themselves in their uh expected States
uh or more precisely their expected
phenotypic and ontogenetic states and uh
consequently they act to minimize the
discrepancy between um the desired or
predicted the State of Affairs uh and uh
the one that they experience so uh uh to
Cote frisen himself uh it furnishes an
account of basic or biotic sentient
behavior that places agency Center Stage
so uh I believe active inference and Fe
uh is not a
deterministic uh theory of cognition at
all uh because uh as I just quoted
friston it takes agency very seriously
and uh even at the very central stage of
the
theory thank you Oli
Susan I kind of toe on that and you
and pose a question in terms of agency
um I you know I kind of question whether
we Rec we recognize without learning our
requisite limits which is kind of what
you I interpreted what you were talking
about um you know people go over their
requisite limits all the time it's
called anxiety it's called you know
schizophrenia whatever but but it's this
has kind of got me thinking that the
agency and the ordance is is kind of a
chicken and egg problem it's like you
know as I as I
update um is I find more freedom um
which you know freedom to choose and in
fact you know when I heard the intention

I was like well I have a choice of other intentions so that's going to Cascade down um that's you know that's basically agency but but I still have to so the belief updating or learning um you know is going to show me new affordances but then I need new agency to be able to do something else with that affordance so you know so is it kind of a chicken and egg problem or how does that play out in this model a few narrow points but first Ali please uh well yes actually uh there is some concept overlap uh the FP uh with the kind of ecological psychology uh Viewpoint offered by Walsh in terms of affordances so uh Walsh argues that uh if an agent has a goal it will experience its environment as a kind of menu of affordances uh what's in its surroundings and will and uh whether uh they're useful or not for attaining that particular goal so uh the environment agent interaction might might be regarded as a kind of landscape uh that uh that should be navigated to that given end uh so in effect what it says is uh it it posits a kind of a optimization problem uh in which um one might effect efficiency um uh to play a role in determining that trajectory so that's basically uh the overlapping between the formulation of

Fe and uh this uh psych ecological
psychology Viewpoint or at least Walsh's
formulation of eonian affordances
because uh in both of those um cases uh
there there's this bilateral uh
interaction between
um between the environment and the agent
so as to uh the chicken egg problem uh
it cannot really be separated from each
other so uh there's this continual um a
bilateral interaction between those
two Co yeah just just to narrower points
so Pablo kind of kicked the soccer ball
off with this question about bottom up
and top down and uh these generative
models are as constructed so we could
imagine a top down constraining Force
that's just always doing it's always
clamping the same way or it's
alternating between two different top
down impositions so just the the the
mere presence of top down
forces does
not support or
reject in the system of Interest
itself the presence or absence of Free
Will defined as as someone chooses um
but by comparing a portfolio of models
it might be possible to to Garner
evidence or kind of um come to favor one
of those choices more and similarly the
um information coming bottom up
could be generated deterministically it
could be an oscillator or it could be
generated probabilistically like being
drawn from a
distribution um and that could drive
belief updating in the agent in a

nonlinear way or in a linear way so it's like sometimes these um adjectives that modify the probabilistic internals of the model deterministic probabilistic all of that um there's there's a lot of richness but also openness in how they articulate with um the broader philosophical questions like Robert Spolsky's current book or it's coming out in some days or something like that it's called determined the science of behavior or living without Free Will so it's like deterministic sounds like it's going to be addressing exact same question that Spolsky is going to be addressing but then as Kristoff pointed to even a deterministic system of quite limited sophistication the only way to predict its behavior may be to play it out this is kind of like Turing's halting problem like even if you're an expert in programming you look at a piece of code you may not be able to know a priori whether it's going to halt or how it's going to act um and then again to this kind of like more limited generative models have a zone of surprisal around them but the zone of surprisal with respect to the modeler isn't the same thing as a truth claim about the system you could make um an example where the affordance landscape is fixed like the rat in the t-maze that we'll see later but a real rat in a t-maze might discover a new affordance like it might

find out it could roll over and you know
knock something over and then if it has
that capacity to
learn that could be model as like
expanding its e variable like expanding
its um repertoire of
affordances and then it might come to
select those so then it but the fact
that a given generative model does or
doesn't have the ability to expand its
repertoire of affordances
the real rat on the
ground just because the model can't
doesn't mean the rat can't just because
the model does doesn't mean the rat will
so that's the kind of iterated continued
engagement
and repeated modeling as a way to
develop hypotheses to to reenter the
system of Interest with action not to
just make version one of the generative
model shut the door on the
question cool
well what's another question or section
that somebody wants to look
to I'd really like to capture what you
just said about the
rat
that was Pro
prolific oh mammals um let's look at
some classic
questions there's a lot here but does
anyone want to just ask one just that
comes to mind now or something they
scribbled in a margin um yeah inks and
then oh yeah inks first and then Andrew
y
so considering the difference between

model and process I'm still very far
behind all of you guys so I was
thinking the model doesn't have to be
true right it's subjective it has to
work it doesn't have to be true does the
process have some limit like do we know
absolute truth on the X asterisk the
hidden state that is asterisk as part of
process I'm thinking that connecting
that to

um I don't know if you guys have read
Greg's Egan uh Greg Egan's Shields
ladder where they discover a v well by
mistake through an experiment they
trigger a vacuum that is a more stable
State than our vacuum and it
starts unrolling through the universe
the way Ice Nine does in cat's cradle by
Kurt Vonnegut so I was thinking our
mod like basically inventing more and
more stable models they roll out like
that like a carpet through everything
because they're so much easier to
survive and persist
through but that's on the process side
the real real hidden state does it have
truth

yeah that that reminds me of this very
short boris um story where um you know
this is this is a somebody read chapter
6 little bit too literally um and they
wanted a perfect
map and then it it came to just like you
said inks it came to unfurl over the
whole
territory until it just blurred with
with the
territory another answer to that is um

xstar the the generative process using the way that the textbook talks about it though Oli may may bring up a very Salient point is um still something that's proposed by the model and so xstar could be said to have like a local truth let's just say we have um you know an air condition turning on or off agent kind of regular setting like that thermometer sense readings making inference about temperature in the room taking action to turn on the air conditioner and then we have like the generative process which is giving rise to the temperature readings now all of that is like a local constructed truth but just making some sort of function to describe how temperature um plays out in that um modeled room again that doesn't mean that the real room has that feature

Oli uh well yes there so there's actually this um kind of criticism or uh debate going on around FP um about whether it's uh too General to be uh realistic uh scientific model of uh the agents or more specifically cognitive agents because on one hand um it kind of puts everything in a a great degree of abstraction uh to the extent that it may not be able to answer some uh some concrete um questions about the biological or the bi olical nature of the uh agent or the real world situations of the agent so for instance

uh there was this a paper by uh coloman
palasio from uh

2020 one uh

namely uh let me yeah non-equilibrium
thermodynamics in the free energy
principle in biology uh in which um they
explicitly put forward this criticism
around FP as being too General to be
useful for modeling any real world or or
concrete uh biological problems but on
the other hand uh FP never claimed
otherwise because uh according to the fe
uh any or a given non-equilibrium system
uh with a Target State um will all would
always seek the most efficient route uh
and this route would typically uh be the
one

defined by considerations of uh the
gradient descent or in other words the
path uh through all the available
configuration space um uh that follows
the uh steepest route uh to the minimum
so uh again in this view uh it's it's
not

even uh explicit whether the agency or
all the other Concepts we're talking
about are real or or the apparent
properties of the agents uh

so um more to the point uh I believe uh
Fe and active inference can be more
congruent with a structural realism uh
view of the uh philosophy of science as
opposed to just um old-fashioned or
traditional uh real pure realism or
naive realism of U view of the science
uh and in this regard uh Frist

coauthored some papers with M Majid
Benny

uh and they argue for this view of f as

uh as a kind of view

that that is more congruent with a

structural realism and particularly

ontological uh structural

realism awesome

Kristoff you know I just want to say

that um holding on to any kind of notion

of Truth U I think the the best the best

thing that ever happened to me in my

life is I decided that just there is

there are no truths there are just

simply models within models within

models and you can sort of see it in in

the temperature example temperature is

not real it's a virtual variable

literally what what the physicist would

call it right it's a um it's a way of

cor spraining all the um interactions

between gas molecules in such a way that

we that we can measure so we have

something that persists for time that we

we can observe but you're you're never

really looking at reality and if you if

you go down you know this rabbit hole

then ultimately you will arrive at

something that like there is some

computational say W's perspective on

this am very fond of that one there is

some computational model that's running

all possible computations so what truth

would be in that in that kind of context

would be that the the the shortest

description um the shortest possible

program that gives rise eventually to

what what you're observing but because

un because Universe because everything

is kind of like interdependent you're

never going to be able to to to to I
think I think you're never going to be
able to separate um you know and say
like oh this is this is the the the the
the the true process that gave rise to
to to this observation it's going to be
some kind of interplay of all kinds of
computational process
all all running kind of independently
so um you know in so far as I think this
is why there there's this difficulty in
in writing down these generative models
of all these possible um Poss when we
have this um you know probability of of
our priors because there's absolutely no
way to to to know what these priors will
be we can only make um uh you know
guesses of what the what the range of
these um uh values uh can taken and
adapt as as um evidence evidence as
observations are flowing

in

cool also on this kind of generative
model generative process uh point this
is one great great
question

um in figure

22 and throughout this

textbook pains are made to distinguish
the generative model from the generative
process and it is a useful distinction
to understand the process that gives
rise to the

observations the modeled process that
gives rise to the observation so the
kind of like digital niche of this
generative model of the agent um and yet
in a guest stream we saw a slight

different perspective or at least way to talk about this when Maxwell presented and utilized this bracket to cover the entire generative model including external

States again those that give rise to sensory States and at this time step in the published transcript Ali asks this question very specifically so there's of course many ways to speak this or any dialect but within this textbook generative process is used to refer to the process external to the agent the particular states of the agent blanket and internal the external states are described as generative process giving rise to the observations and not influencing the actual fundamentals of the particular partition but rather just using a different natural language to describe it it is also becoming common to describe the generative model as describing all the necessary and sufficient statements that the modeler makes about the entire scenario and not distinguishing so much as to what's the generative process in the generative model one piece of uh I guess support for like this kind of just use generative model to describe it all first off it's simpler also if you have two agents in conversation then of course one of them is the generative process to the other and vice versa and so just In The Same Spirit of

like well states of the world or aspects of the world let alone the map they're not tagged as like sensory or action or internal or external those qualifiers are always relative to which node is being called the internal State and then the rest of the blanket is just trivially assigned from there so it doesn't make sense to like say well this desk is the external State it's like well but it's its own internal state if I chose to model it from that side or with respect to the wall behind it it's the blanket State between me and the wall

so using generative Model A little bit more

broadly seems to be a little bit of a development that has unfolded over the uh cohort and years

Susan yeah I mean the way ask see it if I you correct me if I'm not looking at it correctly but you know when you do research you have to decide what your ontology or epistemology is going to be when you apply it um and um and you always provide context for that so this kind of is above that in my mind in fact you know we and you and I've had a conversation because I yeah I want to make sure that the that it's um that it's ideology free

um

because you know I see way too much um you know um BS especially in management science that you know is just embedded with somebody's ideology and it's taken as fact so it's kind of what I like

about the model um that um brings me to
um you the the marov
blanket and um you where you so in that
sense of you the uh the model the
generative model where where does the
Markoff blanket extend because it was a
little confusing reading
it is that just confirm for me though am
I looking at it correctly in terms of it
being a
theistic that's why I posted that in the
comments because that's what I copied
from chapter 10 I think
cool maybe post what what page you you
found it on but it's it's a great quote
um features are not intrinsically on the
blanket or internal or
external there's a base graph that
conveys whatever variables are being
included
whatever Quantified aspects are under
consideration or or formalized doesn't
have to be a number it could be like
categorical but whatever is being
brought whatever Lego pieces are coming
on the table and how those Lego pieces
are connected that's the base graph and
then whatever node or cluster of nodes
is being selected as the internal States
the blanket is just all that surround
that and so when we're taking this kind
of agentic view or cybernetic view on
cognitive ecosystems then
the blanketing states just to give a
first pass can be understood in terms of
sensory data so anything that you would
record with a sensor and then the
outgoing dependencies are actions

anything that would be described by the affordances of an actuator so to add so to add build on what Ink's question was would the Markoff blanket be the same for if I'm doing let's say I am doing a research in a modeling project would I use the same Markoff blanket for the process and the model or would the model or would the process change the blanket the the final mod just giving a narrower answer because I I I know that's something that could be explored at Great depth if you made a generative model with a fixed Mark marov blanket then it would be fixed if you said our room has a thermometer and an arm that's an actuator and you know different data are going to come in and the arm is going to do different things well then the the kind of um scheme of the blanket wouldn't change but you might want to be interested modeling the transition from that room deciding that it should put out a second thermometer at which point the scheme of it would change okay and so then you'd be outside that room be like oh they literally just went from having one thermometer to two thermometers and then you could postulate cognitive models that happened inside of that room you'd say maybe it's like there they had an affordance to add a thermometer and they selected that affordance does that answer your question better

in now I have more more questions but I
don't want to stir more trouble
here what what else is the end of a
session for other than stirring trouble
um so because you were talking about
being in the room can I just roll with
it of

course you were talking about being in
the room and then a second th
thermometer and I was thinking of as an
agent this I don't think this is about
this chapter but as an agent as my mark
of blanket is like I'm looking at the
screen right now but then have a flutter
in my heart and I'm like oh no am I
having a heart attack in my mark of
blanket is that a separate Mark of
blanket where now I'm modeling myself
very close and tight around my heart
then I hear the door knock and oh I have
a package from Ukraine and I run like
now my My atten sl Mark of blanket is
all the way my apartment and the door to
the outside world where someone is
knocking are these different am I
switching

between different Mark of blankets or is
the mark of blanket moving with my
attention

regime great question well to the
wandering blankets and um blankets that
are not sort of one or zero but can be
on a Continuum or a blanket index
certainly some of the most recent work
is addressing that I'll just leave that
there but to the interception question
um in chapter six when we're exploring
ing how to construct these generative

models an example of of Sensor Fusion is provided so let's just say that a given body has some interceptive Sensations like heart rate and then some exteroceptive sensory capacities like hearing and vision and so on um those are kind of like all open even when you close your eyes you're still getting visual input and and usually it's modeled not as changes in the blanket but in changes in attention like you still have proprio receptors in your foot but you're not paying attention to them you're not attending to them which is to say it's as if this the data coming from them are not making a difference to belief updating if you're not paying attention to something it is not having consequence it's its differences are not making a difference and then regimes of ention describe the way that in this kind of multimodal setting um the spotlight of attention might focus on a given sensory modality and so that is usually modeled more as an attention modulation task within a fixed marov blanket schema rather than um changes to the the schem of the blanket itself

Kristoff so you know this makes me think that um having this sort of uh you know explicit model of attention within um a fixed Mark of blanket it's really it's really Mark of blanket is a kind of like a convenient um it's it's an idea it's a convenient

idea for modeling to to make this separation but it's not actually something that that is real there is um any boundary that you can find and say like yes this is definitely a boundary between the internal and external States and this recent work on on on on moving blankets I think I think what we're eventually going to realize is that it's there's just a Precision parameter there as well like how how how how thick I want my my market blanket to be and um it's for for the convenience of describing certain phenomena at certain temporal scales it simply makes it makes sense to make an assumption that there is a separation between these internal and external States but in reality if you look at it you know at all temporal scales you will find that there is the the the separation is you know it's not really there it's a it's a it's um it's it's invented so this you know it's kind of like uh this is the way I sort of understood the the the Buddhist concept of relativity everything is sort of relative and I I put myself in a frame of reference and I said well within this frame of reference these are internal these are external States and then I have and and and I have a way of looking at them to make sense of them if if I didn't have the ability then it might be difficult to to be able to make sense sense of the world if I if I if I can't make this partition but is the partition real I I you know

personally I didn't think so it's just

it's just a point of

view thank you um

Andrew

yeah a couple of riffs on um what's been

said uh one like the the the finding

your own Markov blanket I feel like is a

little bit of a um hopeless cause

because it it's it's very likely that

you are many

nested um you contain Markoff blankets

all the way down um and

there's um a lot of complex phenomena

and interior modeling going on um in

terms of coming to grips with that one

resource I'd like to recommend is if

anyone's seen the book The Mind

illuminated by Dr John

Yates um it's it's kind of a good

rationalist take on a meditation

practice that uh kind of lets you come

to grips with there's probably multiple

agentic things happening in your brain

um another thing that uh was just kind

of wow to me that I just wanted to gush

about about this chapter is um I've been

trying to sort of come to grips with

what normativity means in a um sort of

non-dist agentic um frame of reference

and there was a couple quotes at the end

of this chapter one of which is in

active inference the identity of an

agent is isomorphic with its priors that

just like I was like oh all right that

kind of puts it all together and um kind

of um to to get that um pitched up they

start with expected states are preferred

and include the organism's conditions

for survival EG Niche specific goal

States whereas their opposite surprising

states are dispreferred in this way by

find fulfilling their expectations

active inference agents ensure their own

Survival so it just that phenomena one

thing

um your Beijing priors are your

normative order which um I've been

searching for what does a normative

order mean especially within the context

um sort of

a small agent um trying to find

relevance in in the cognitive limits of

finding relevance there and um I feel

like this is um this is the right

approach also one more

thing with the um free uh not

variational free energy but um um the

the planning one expected free energy

um we were talking about um what s the

way it includes both

epistemic and um preference value in the

same equation I I feel like as a group

we really need to um tackle that as a

and realizing that maybe a lot of

our preconceptions that we've been

grappling with during this meeting are

really summed up with understanding that

itself that that it all fits both

epistemic ontological in the same sort

of optimization

framework so that's that's just me um

riffing on this great comments equation

2.5 and equation

2.6 are like the heart of the heart of

active inference we have the real time

sense

making about incoming sense data and beliefs and then we have the prospective which not all agents even need to be modeled as doing the prospective unified imperative for pragmatic and epistemic outcomes and that's the heart and that's the mystery and that's um like the part that we don't need to reinvent and then all the work is in getting it into a form getting the system of Interest into a form where you just press play on those equations um Ali and then any final comments that people have thank you uh yeah uh regarding whether or not or to what extent markup blankets are code UN code real uh it's probably worth uh keeping in mind uh that markup blankets are uh more precisely they're not necessarily spatio-temporal boundaries but rather uh they draw boundaries in state space or in other words in the space of all possible configurations uh so uh it's true that sometimes it manifests itself as a kind of spatio-temporal boundary such as uh the cell membrane uh but uh more often than not uh we don't need to consider markup blankets as uh kind of purely physical or spatio-temporal boundaries so uh yes it's a kind of uh modeling technique uh but but uh in terms of its mathematical um reality it is as true as

any other mathematical
models cool
well very fun the low
road is it a bumpy road is it a smooth
Road what is it like I don't know um but
that was our first discussion on it next
week at the earlier time we will check
back so it would be epic if people just
right in the coming minutes or in the
coming week add a few more questions to
chapter five questions where we'll look
at the bigger questions table too um in
cohort 4 we'll be heading into chapter
eight on continuous time generative
models and uh we'll have next week on
chapter two then head into three so
thank you all the game goes on so see
you next
time thanks everyone bye bye